

Exam project work

Tecnologie e Applicazioni Web / Web Applications and Technologies, a.y. 2025/2026

This year's project involves developing a web application comprising a server with RESTful APIs and a single-page application (SPA) front end to manage overseas mobility at our University¹.

The Overseas program allows Ca' Foscari students to undertake a period of mobility at a non-European partner university. During the mobility period, the student can attend courses and take exams at the host institution. At the end of the stay, exams successfully passed abroad can be recognized within the student's study plan at Ca' Foscari. The goal is to create a web application to simplify the management of the main administrative phases of overseas mobility: before departure, during mobility, and after return.

The system must manage at least three types of users:

1. **students**, who create their mobility applications and upload the required documents;
2. **referent lecturers** (docenti referenti), who evaluate mobility applications and handle the recognition of exams;
3. **overseas office staff**, who monitor the applications and close them at the end of the process.

A student can create one or more mobility applications. Each application refers to a single host institution, chosen from a predefined list of partner institutions, and a single referent lecturer at Ca' Foscari.

Procedure workflow

The overseas program involves multiple steps.

Step 1: Before departure

The student creates an application by entering a few essential details:

- academic year;
- host institution;
- expected mobility period (first semester, second semester, or the entire year);
- referent lecturer;

In addition, the student must enter the mapping between the exams he or she intends to take abroad and those of his or her Ca' Foscari study plan, indicating:

¹ Note: the project goal is the same as the one proposed in the databases courses, but specific requirements may vary. Contact the teacher for any questions.

- foreign teaching code;
- name of the foreign course;
- credits for the foreign course;
- course code in the study plan at Ca' Foscari;
- title of the course in the study plan at Ca' Foscari;
- teaching credits in the study plan at Ca' Foscari;

After creating the application, the student also uploads a *Learning Agreement* (see Attachment A), which is a formal document describing the student's application. For the purposes of this project, the *Learning Agreement* can be managed simply as a user-uploaded file; it is not necessary to model the document content.

The referent lecturer must be able to view the application and the document, and approve or reject it. In case of rejection, they must be able to provide a reason.

Once the student application has been approved by the referent lecturer, the Overseas Office verifies the pre-departure application and registers in the database that the pre-departure phase is complete.

Step 2: During mobility

The student indicates the actual arrival and departure dates at the host institution. If the exam schedule changes, the student can propose a modification to the list of exams planned abroad and their relative mapping, together with the upload of an updated *Learning Agreement*.

The referent lecturer can approve or reject the modification directly from the application interface. In case of rejection, the original exam mapping and *Learning Agreement* must be restored.

Step 3: After returning

After returning, the student uploads the Transcript of Records, which is the document issued by the host institution listing the exams taken abroad and the results obtained.

Based on the Transcript, the student also updates the list of exams with the score obtained and the date on which the exam was taken.

Then the referent lecturer reviews the data and approves the exams along with their scores.

Once this phase is complete, the Overseas Office can verify the application and close it.

Mobility applications

Each mobility application is created by a student and then possibly modified by referent lecturers or Overseas Office staff. Each application must have an overall status, for example:

- created;
- awaiting Learning Agreement approval;
- pre-departure completed;
- mobility in progress;
- waiting for exam score approval;
- closed;
- canceled.

Architecture

The system must comprise:

- A REST-style **backend** web service, implemented in TypeScript or JavaScript, and running on Node.js. The backend must use Express.js for routing. For the database, you can choose **MongoDB** (as seen during the course) or a **Relational database**.
- A web **front-end** implemented as a Single Page Application (SPA) using the Angular framework.

Note: Each component (Backend, Database, Front-end) must run in a separate Docker container.

Requirements

Your application must implement at least the following features:

User and Role Management

- The system must distinguish between students, referent lecturers, and Overseas Office staff.
- A student can only view and modify their own applications.
- A lecturer can view the applications for which they are the referent.
- The Overseas Office can view all applications.

Host Institution Management

- The system must contain a predefined list of partner institutions.
- Each institution must have at least a name, country, and city.
- The student must choose the host institution from this list.

Creation of Mobility Applications

- A student must be able to create one or more mobility applications, specifying the academic year, host institution, expected period, referent lecturer, and the mapping

between the exams he or she intends to take abroad and those of his or her Ca' Foscari study plan.

Learning Agreement Upload and Evaluation

- The student must be able to upload a Learning Agreement file as part of an application.
- The referent lecturer must be able to approve or reject the Learning Agreement and record the date of the decision and the reason.

Pre-departure Verification

- The Overseas Office must be able to set the pre-departure phase as complete only if the application contains the essential data and the Learning Agreement has been approved by the referent lecturer.

In-Progress Mobility Management

- The student must be able to enter the actual dates of arrival and departure at the host institution.

Learning Agreement Modifications.

- The student may propose one or more modifications to the exam mapping during the mobility period. Each modification contains a textual description. Such modifications imply uploading a new Learning Agreement.
- The referring lecturer must be able to approve or reject each modification, recording the date and, if applicable, a reason.

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Transcript of Records Upload

- After returning, the student must be able to upload the Transcript of Records related to the application.
- The referent lecture reviews the Transcript of Records and approves the exam taken, along with its score.

Application Closure

- The Overseas Office must be able to close an application only when the Transcript of Records has been uploaded, and the referent lecturers have approved all the exams.

The project description is deliberately presented at a high level and lends itself to developing projects of varying complexity, depending on the group members' tastes, imagination, and abilities. Consider the outline a simple starting point; you can use your creativity and experience with similar applications to identify new, interesting features to implement.

Some possible ideas for enriching the scenario might be:

- Provide a history of files uploaded as Learning Agreement and Transcript of Record, with the upload date of each version.

- management of document signatures via download/upload, defining which users must sign the document and the order in which they must sign, and tracking the corresponding progress status.
- dashboard for the Overseas Office with applications by status, country, or host institution;
- notifications for incomplete applications;

Additional requirements:

- When the backend starts, it must preload test data into the database to simplify application testing during the exam.
- User authentication must be implemented using JSON Web Tokens (JWTs), stored either in cookies or in the browser's local storage.

Submission

Projects must be submitted via Moodle at the following page:

<https://moodle.unive.it/mod/assign/view.php?id=1117362>

Each member must submit the same file, named <group_name>.zip, if working in a group.

The package must contain:

- **The report of each student**, in PDF format, named:
<student_firstname>_<student_surname>_<matriculation_number>.pdf
- A README.txt file with instructions on how to run the entire application. For example, you can briefly summarize the shell commands to compile the sources, start the backend and frontend, etc.
- All the application source code, clearly divided into backend and frontend.

NOTES:

- Do not submit any external libraries. Please delete all the node_modules directories before submission.
- Students can freely form **groups of up to 4 people**. Moodle will not help or enforce group submission. Individual students can simply use their surname as their group name.

Report

A report describing the system architecture, the software components used, and how they contribute to achieving the required functionalities must be submitted with the application. The report is **strictly individual** and will serve as a basis for the oral discussion of the project.

The report must contain:

- A description of the system **architecture**, listing all the different components and their relationships (i.e., how they work together to implement the application requirements).
- A description of the **data model**, either the ER diagram if you've chosen a relational database or the list of collections and the documents inside each collection if using MongoDB.

- A precise description of the **REST APIs**. Such a description must include the list of endpoints, parameters, and the data exchanged. Provide examples in JSON format whenever possible.
- A description of how user **authentication** is managed, along with the associated workflow.
- A description of the Angular front end, with a concise list of the implemented **components, services, and routes**.
- Some examples, possibly with **screenshots**, showing the typical application workflow for each different user role.
- Description of what and how AI tools were used and the results obtained.

AI

AI tools can be used to help with development, but you are fully responsible for the code produced. During the oral, I may ask you to describe, modify, or write new code for your project without any AI assistance.

Q&A

For any questions about the project work or the course in general, don't hesitate to contact the teacher at:

filippo.bergamasco@unive.it

I'll be happy to arrange a meeting if needed!

Additional information about the exam is available at:

<https://moodle.unive.it/mod/page/view.php?id=1080337>

Prof. Filippo Bergamasco,
Tecnologie e Applicazioni Web, a.y. 2025/2026

Attachment A: Learning Agreement

Here is an example of the *Learning Agreement* describing the student's application. The same data should be managed by the application.



– International Office, Ca' Foscari, Dorsoduro 3246, 30123 Venice, Italy –

Università
Ca' Foscari
Venezia

International Exchange Outgoing Student

LEARNING AGREEMENT

Please fill in the form and send it to the International Office of Ca' Foscari University of Venice by e-mail (overseasout@unive.it).

DETAILS OF THE PROPOSED STUDY PROGRAMME ABROAD/LEARNING AGREEMENT

Academic year	_____	Semester	_____	Matriculation number	_____
Student's name	_____	Student's surname	_____		
Department	_____				
Sending institution	Università Ca' Foscari Venezia			Country	Italy

Hosting Institution _____ Country _____

Course unit code	Course unit title	Number of university credits
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC
OC	OC	OC
EC	EC	EC

OC = Original Course at the hosting university; EC = Equivalent Course at Ca' Foscari University.
If necessary, reprint the form and continue the list.

Student's signature _____ Date (dd/mm/yyyy) _____

SENDING INSTITUTION:
We confirm that the proposed learning agreement is approved.

Academic contact person's signature _____

Date (dd/mm/yyyy) _____

HOSTING INSTITUTION:
We confirm that the proposed learning agreement is approved.

Academic contact person's signature _____

Date (dd/mm/yyyy) _____